



**TEMASEK JUNIOR COLLEGE
PRELIMINARY EXAMINATION
JC 2 2018**

CANDIDATE
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H2 BIOLOGY

Paper 2 Structured Questions

9744/02

**Friday 24 August 2018
2 hours**

READ THESE INSTRUCTIONS FIRST

Write your name, Centre number, index number and class in the spaces at the top of the page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graph.

Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use					
Q1	/ 8	Q5	/ 10	Q9	/ 12
Q2	/ 11	Q6	/ 10	Q10	/ 8
Q3	/ 11	Q7	/ 5	Q11	/ 8
Q4	/ 7	Q8	/ 10		
Total					/ 100

This document consists of **27** printed pages and **1** blank page.

Answer **all** the questions in this section.

1 Plants vary greatly in terms of size.

(a) Explain whether the cell theory is applicable to plants.

[2]

Sugar molecules enter cells through transport proteins.

(b) Explain why transport proteins are required for the movement of sugar molecules, such as glucose and fructose, into cells.

[2]

Some plant cells convert fructose and glucose into sucrose for transport from the leaves to the roots. Sucrose is moved into phloem sieve tubes as shown in Fig. 1.1.

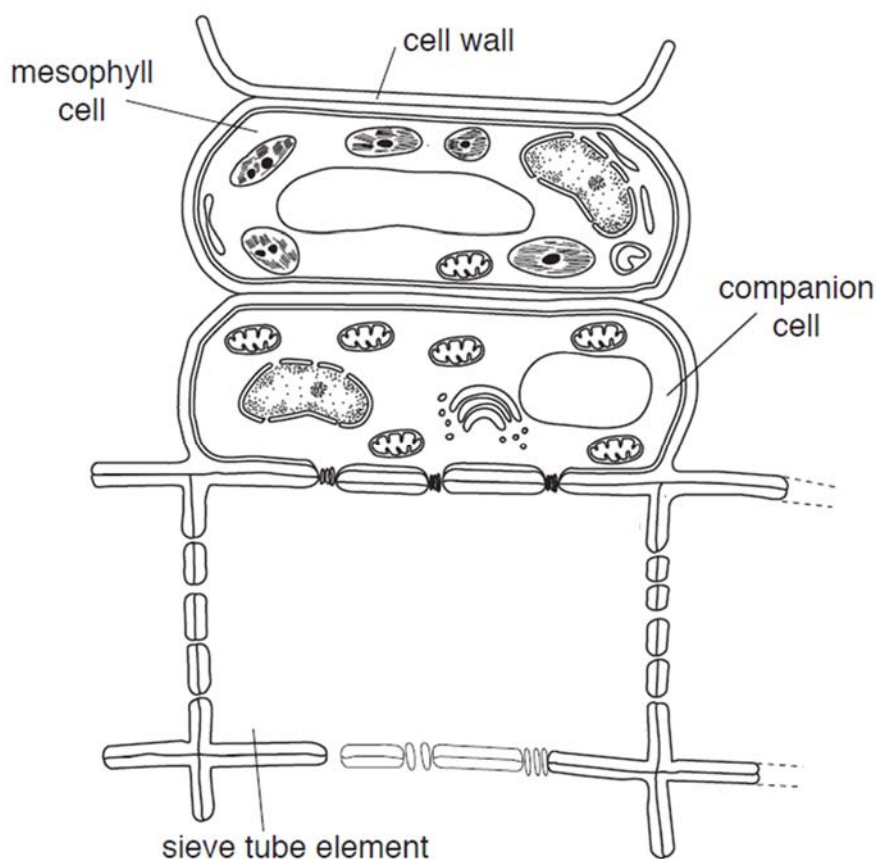


Fig. 1.1

Each cell has a specialized function.

- (c)** With reference to Fig. 1.1 and the information provided, state **one** difference between a mesophyll cell and companion cell.

[1]

Fig. 1.2 shows how sucrose is transported into the companion cell from the mesophyll cell.

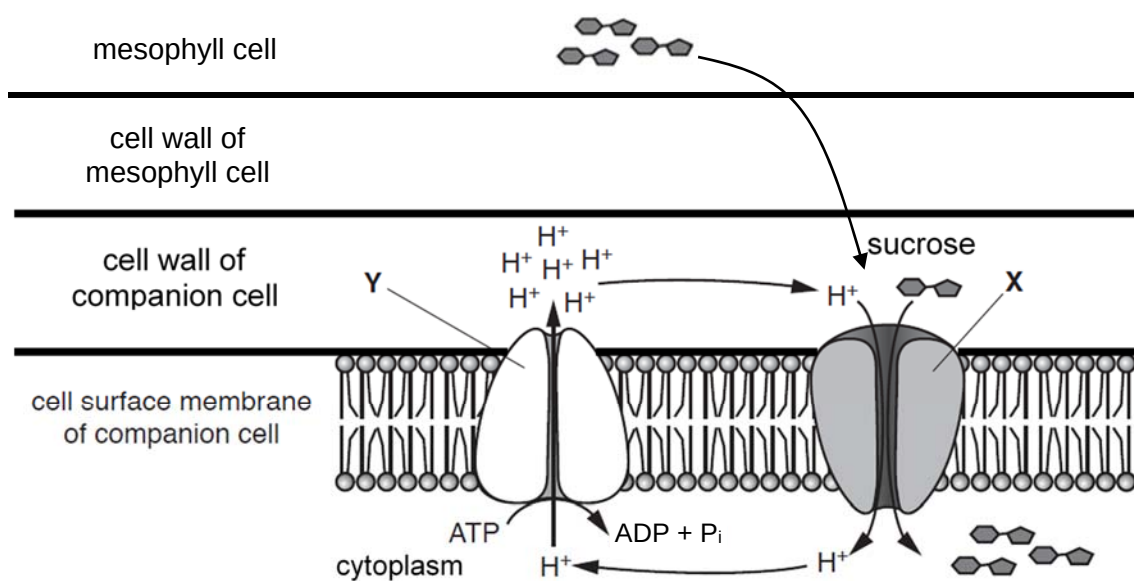


Fig. 1.2

(d) Using the information in Fig. 1.1 and Fig. 1.2, explain how sucrose moves into the companion cell.

[3]

[Total: 8]

- 2 The yeast, *Saccharomyces cerevisiae*, is a single-celled, eukaryotic organism that is often used in the laboratory.

When yeast is mixed with a glucose solution, the yeast absorbs the glucose. Each molecule of glucose is then broken down into pyruvate molecules in exactly the same way as in any other eukaryotic organism.

- (a) Outline the breakdown of glucose to pyruvate in this stage.

[2]

Yeast cells sometimes carry out anaerobic respiration. Fig. 2.1 outlines the process of anaerobic respiration in yeast cells.

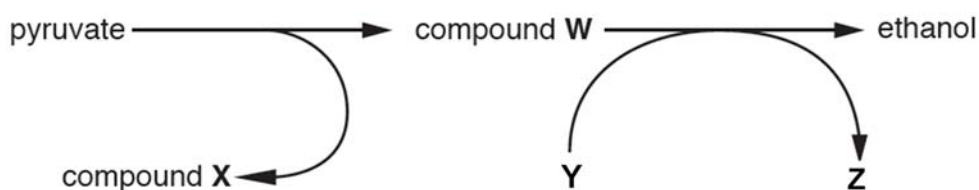


Fig. 2.1

- (b) (i) Identify molecule Z.

[1]

- (ii) State why molecule Y is converted to Z.

[1]

Table 2.1

Time / min	Height of dough / cm		
	Fridge	Room temperature	Next to window (hot day)
0	2.5	2.5	2.5
20	2.5	2.9	3.3
40	2.7	3.7	4.0
60	2.9	3.9	4.7
80	3.0	4.0	5.2
100	3.0	4.0	5.8
120	3.0	4.0	6.0

-
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-
-
-
-
-
-
-
-
- [4]

- (ii) Suggest why the increase in the height of dough that was placed at room temperature was higher between 0 and 40 minutes than between 40 minutes and 60 minutes.

[2]

- (iii) Suggest why the height of the dough that was placed at room temperature ceases to increase after 80 minutes.

[1]

[Total: 11]

- 3 In maize plants, a gene locus for leaf colour and a gene locus for cob colour were studied.

A pure breeding maize plant with bronze leaves and brown cob was crossed with a pure breeding maize plant with green leaves and yellow cobs to produce F1 phenotypes.

All the F1 plants had bronze leaves and brown cobs.

- (a) Define the term *locus*.

[1]

A test cross was conducted for these two loci using the F1 plant. Table 3.1 shows the results of this cross.

Table 3.1

Phenotype	Observed number (O)
bronze leaves and brown cobs	44
bronze leaves and yellow cobs	6
green leaves and brown cobs	7
green leaves and yellow cobs	43
Total	100

- (b) (i) Use the symbols:

L bronze leaves; **l** green leaves; **B** brown cobs; **b** yellow cobs

State the phenotype of the test cross plant.

[1]

(ii) Draw a genetic diagram to explain the results of this test cross.

[4]

Another type of maize plant produced a total of 381 grains, 216 purple and smooth, 79 purple and shrunken, 65 yellow and smooth and 21 yellow and shrunken.

A chi-squared test was carried out to test the significance of the differences between the observed and expected results.

Table 3.2 shows some calculations to obtain the chi-squared value.

Table 3.2

grain phenotype	observed number	expected number	$\frac{[\text{obs no.} - \text{exp no.}]^2}{\text{exp no.}}$
purple and smooth	216	$381 \times 9/16 = 214$	$4/214 = 0.019$
purple and shrunken	79	$381 \times 3/16 = 71$	$64/71 = 0.901$
yellow and smooth	65	$381 \times 3/16 = 71$	$36/71 = 0.507$
yellow and shrunken	21		
Total number	381	χ^2 value	

(c) Complete the missing spaces in Table 3.2.

[2]

Table 3.3 shows some critical values for chi-squared test at different probability levels.

Table 3.3

Degrees of freedom	Probability, p						
	0.50	0.10	0.05	0.02	0.01	0.005	0.001
1	0.46	2.71	3.84	5.41	6.64	7.88	10.83
2	1.39	4.61	5.99	7.82	9.21	10.60	13.82
3	2.37	6.25	7.82	9.84	11.35	12.84	16.27
4	3.36	7.78	9.49	11.67	13.28	14.86	18.47
5	4.35	9.24	11.07	13.33	15.09	16.75	20.51

(d) (i) Describe how the degrees of freedom was determined;

_____ [1]

(ii) State the conclusion from the χ^2 value calculated in (c).

 _____ [2]

- 4 Fig. 4.1 shows the effect of increasing substrate concentration on the rate of a particular reaction in the presence and absence of an enzyme.

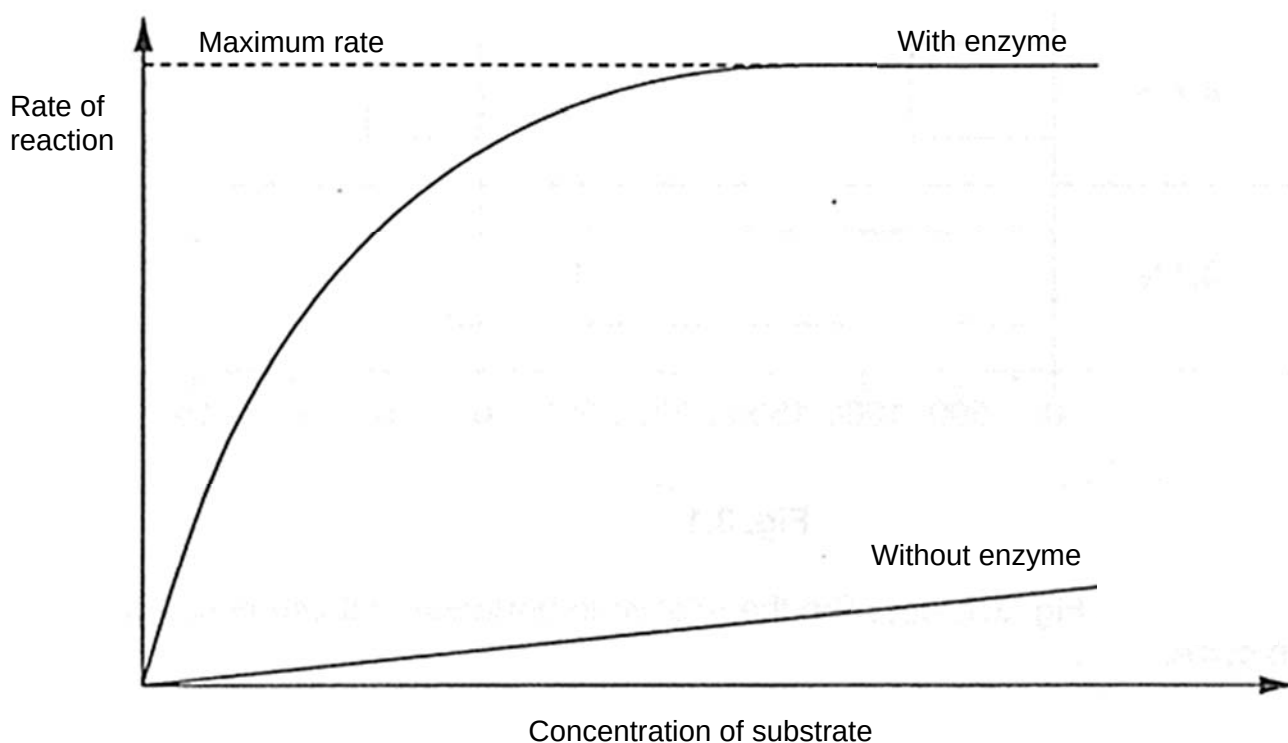


Fig. 4.1

- (a) On Fig. 4.1, draw **two** labelled curves to show the effect on the rate of the enzyme catalysed reaction upon the addition of
- (i) a competitive inhibitor;
 - (ii) a non-competitive inhibitor.

[2]

- (b) Explain the effect of a competitive inhibitor on the rate of enzyme activity.

[3]

(c) State **two** differences between a competitive inhibitor and a non-competitive inhibitor.

[2]

[Total: 7]

- 5 Fig. 5.1 shows a stage in the mitotic cell cycle in an animal cell.

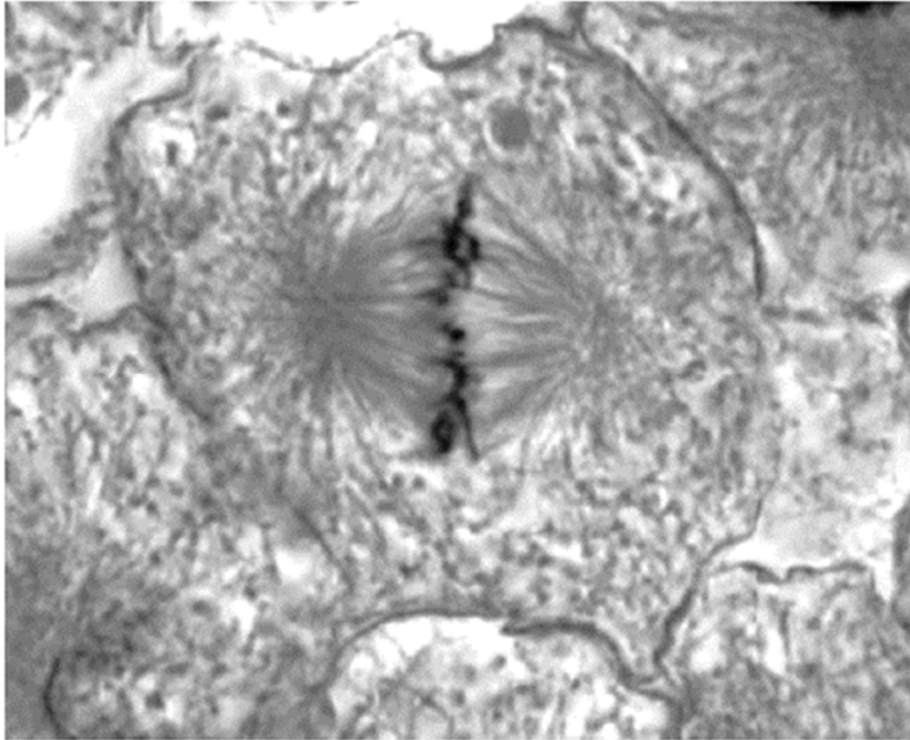


Fig. 5.1

(a) With reference of Fig. 5.1,

(i) identify the stage of mitosis;

_____ [1]

(ii) state two features which are characteristic of this stage.

_____ [2]

(b) Distinguish between the terms haploid and diploid.

_____ [2]

(c) Explain the importance of mitosis in organisms.

[3]

(d) In many multicellular organisms, such as mammals, the time taken for the mitotic cell cycle varies considerably between different tissues, but is very carefully controlled in each cell.

Suggest the importance of this control in mammals.

[2]

[Total: 10]

- 6 *Staphylococcus aureus* is a bacterium that is resistant to most types of antibiotics, such as penicillin.

In a study to understand how bacteria gain antibiotic resistance, a strain of *E. coli* with no known antibiotic resistance was mixed with heat-killed *S. aureus* for 24 hours.

E. coli was then grown on Petri dish containing penicillin and the number of *E. coli* colonies were counted.

For the control, *E. coli* without *S. aureus* was grown on another Petri dish and the number of colonies were counted.

Fig 6.1 shows the result of the experiment.

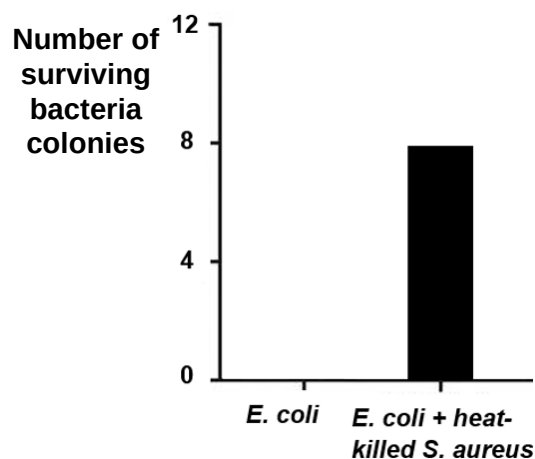


Fig. 6.1

- (a) Identify the process that allows *E. coli* to become antibiotic resistant.

_____ [1]

- (b) With reference to Fig. 6.1,

- (i) describe the results observed;

 _____ [1]

- (ii) Explain the results observed in (b)(i).

 _____ [2]

Fig. 6.2 shows another process by which antibiotic resistance genes can be passed from bacterium **A** to **B**.

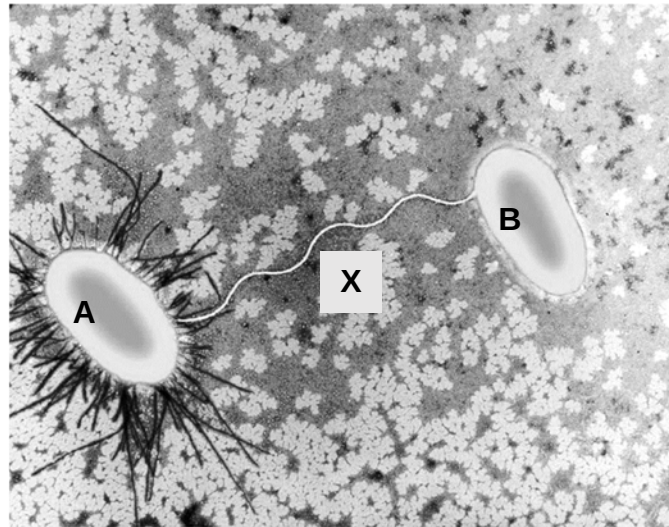


Fig. 6.2

(c) With reference to the process shown in Fig. 6.2,

(i) identify the nature of each bacterium;

Bacterium **A** : _____

Bacterium **B** : _____ [2]

(ii) identify structure **X**.

X : _____ [1]

(iii) Distinguish between the process of genetic recombination stated in **(a)** and in Fig. 6.2.

[3]

[Total: 10]

- 7 In an investigation to study genetic variation, DNA was obtained from four varieties of the same invertebrate species.

The following technique was used:

- DNA was digested using a number of different restriction enzymes to obtain different fragments
- The fragments were separated by gel electrophoresis
- RNA probes were used to select DNA fragments with specific sequences

(a) Explain how RNA probes, used in this technique, select fragments of DNA.

[2]

Fig. 7.1 shows the results after the RNA probes have bound to the selected DNA fragments.

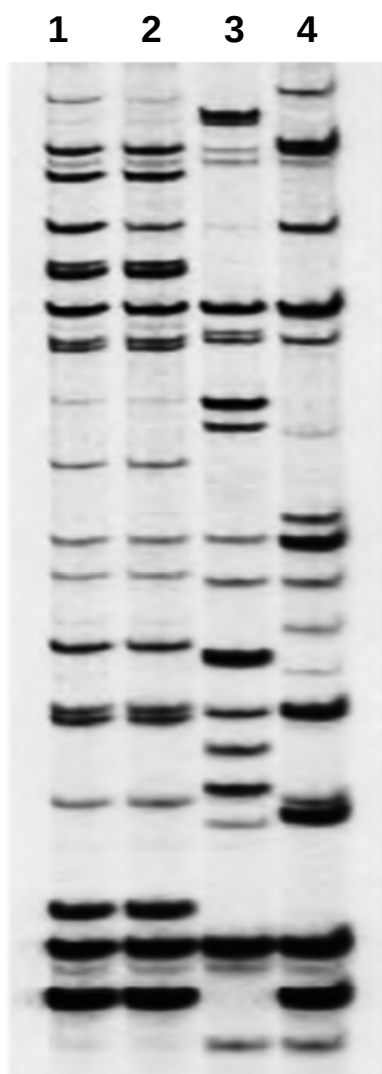


Fig. 7.1

(b) On Fig. 7.1,

(i) draw an arrow (➡), indicate **one** DNA fragment found in all four varieties where the RNA probe has bound; [1]

(ii) Identify the varieties which have the same genetic fingerprint and explain your answer.

[2]

[Total: 5]

- 8 Table 8.1 shows some of the common fatty acids and their melting points.

Table 8.1

Symbol (number of carbon atoms : number of double bonds)	Common Name	Melting point (°C)
<i>Saturated fatty acids</i>		
12 : 0	Lauric acid	44.2
14 : 0	Myristic acid	52
16 : 0	Palmitic acid	63.1
18 : 0	Stearic acid	69.6
20 : 0	Arachidic acid	75.4
22 : 0	Behenic acid	81
<i>Unsaturated fatty acids</i>		
16 : 1	Palmitoleic acid	-0.5
18 : 1	Oleic acid	13.4
18 : 2	Linoleic acid	-9
18 : 3	α -linolenic acid	-17
20 : 4	Arachnidonic acid	-49.5

- (a) Arachidonic acid is a polyunsaturated fatty acid. Explain the term *polyunsaturated fatty acid*.

_____ [1]

- (b) With reference to Table 8.1,

- (i) describe the effect of increasing number of carbon atoms in saturated fatty acids on the melting point;

 _____ [3]

(ii) describe the effect of the presence of double bonds in fatty acids on the melting point;

[1]

(iii) explain the trend described in **b(ii)**.

[4]

(c) Suggest where polyunsaturated fatty acids are usually found in nature.

[1]

[Total: 10]

- 9 Table 9.1 provides statements regarding the bonds found in four biological molecules.

Table 9.1

statement	protein	DNA	messenger RNA	cellulose
hydrogen bonds stabilise the molecule				
subunits are joined by peptide bonds				

- (a) Complete Table 9.1 by indicating with a tick (✓) or a cross (✗) whether the statements apply to proteins, DNA, messenger RNA and cellulose.

You should put a tick or a cross in each box of the table.

[2]

- (b) Telomeres are parts of chromosomes. Describe the function of telomeres.

[4]

(c) A piece of mRNA is 660 nucleotides long but the DNA coding strand from which it was transcribed is 870 nucleotides long.

(i) Explain this difference in number of nucleotides.

_____. [1]

(ii) What is the maximum number of amino acids in the protein translated from this piece of mRNA? Explain your answer.

Number of amino acids _____

Explanation

_____. [2]

(d) Identify **one** other process that leads to the formation of mature mRNA and state its function.

_____. [2]

(e) Describe **one** difference between the structure of mRNA and tRNA.

_____. [1]

[Total: 12]

- 10 Unlike eukaryotes, prokaryotes have different mechanisms for controlling gene expression. Fig. 10.1 shows the Jacob and Monod model of gene expression in the *lac* operon of *Escherichia coli*.

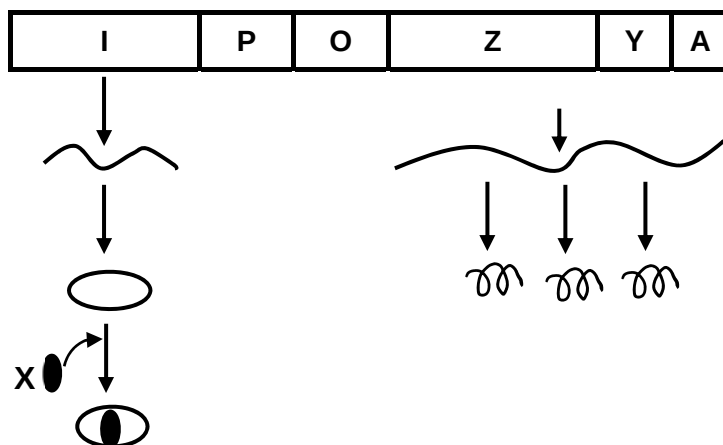


Fig. 10.1

(a) With reference to Fig. 10.1 identify,

(i) identify molecule X;

[1]

(ii) identify region I.

[1]

The regulation of *lac* operon in *E. coli* was investigated and wild-type *E. coli* were cultured in two different agar media.

X-gal was added to both agar media. It is a colourless substance that is converted to a blue compound by the enzyme, β -galactosidase.

Table 10.1 shows results of the investigation.

Table 10.1

Type of agar medium	Colour of colony
X-gal, lactose and no glucose	blue
X-gal, lactose and glucose	white

- (b) Apart from the presence of an inactive *lac* repressor, explain the appearance of the colony when wild-type *E. coli* was cultured in lactose and X-gal without glucose.

[3]

In a separate experiment, scientists fused the *trp* operon with the *lac* operon as shown in Fig.10.2.

The *trp-lac* fusion operon was then inserted into a bacterial cell to replace the separate *trp* operon and *lac* operon such that the transformed bacterium only has the *trp-lac* fusion operon.

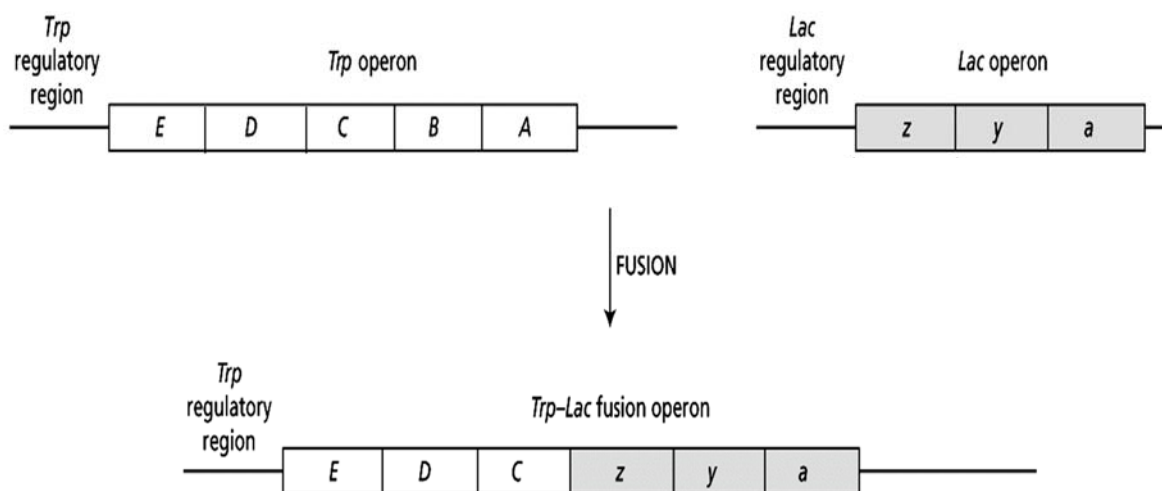


Fig. 10.2

- (c) State and explain the conditions which must be present in order for β -galactosidase to be formed in the transformed bacterium with *trp-lac* fusion operon.

[3]

[Total: 8]

11 Relationships between different primates can be found by comparing their proteins and DNA.

The proteins of different species can be compared using immunological techniques.

The protein albumin obtained from a human was injected into a rabbit. The rabbit produced antibodies against the human albumin.

These antibodies were extracted from the rabbit and then added to samples of albumin obtained from four different animal species. Precipitation occurs when antibodies bind to albumin. The amount of precipitate produced in each sample was then measured and shown in Table 11.1.

Table 11.1

Species from which albumin was obtained	Amount of precipitate / arbitrary units
Rat	23
Chimpanzee	96
Marmoset	65
Trout	11

- (a)** Comment on what the results suggest about the evolutionary relationship between humans and the other species?

[2]

Scientists also used DNA hybridisation to determine the evolutionary relationships between five species of primate.

The separation temperature is the temperature at which a molecule of double-stranded DNA separates into two single strands.

The scientists first recorded the mean separation temperature of DNA in which both strands were from the same species.

The scientists then recorded the mean decrease in separation temperature of DNA in which one of the strands was from another species. Their results are shown in Table 11.2.

Table 11.2

Primate	Mean decrease in separation temperature / °C				
	Human	Chimpanzee	Gorilla	Orang-utan	Gibbon
Human					
Chimpanzee	1.7				
Gorilla	2.3	2.3			
Orang-utan	3.6	3.6	3.5		
Gibbon	4.8	4.8	4.7	4.9	

- (b)** When the scientists first recorded the mean separation temperature of DNA in which both strands were from the same species, differences in the separation temperature was observed. Suggest why this is so.

[1]

- (c)** With reference to Table 11.2,

- (i)** explain if the data suggests that gibbons are most distantly related to humans;

[2]

- (ii) The scientists assumed that the decreases in separation temperatures are directly proportional to the time since the evolutionary lines of these primates separated.

Gorillas are thought to have separated from orang-utans 20 million years ago. Use this information to calculate how long ago the evolutionary lines of humans and chimpanzees separated.

Show your working.

_____ million years [3]

[Total: 8]

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